University Of Asia Pacific

Department of Computer Science and Engineering (CSE)

Course Title: Electronic Devices and Circuits Lab

Course Code: EEE 202

No. of Experiment : 01

Name of Experiment : Design and Verification of RTL Logic Gates

Date of Experiment : 28/07/2026

Date of Submission : 04/08/2026

Submitted By

Name : Utsob Ghsoh

Registration No : 25101003

Section : A1

Submitted To

Abdullah Al Mahmud

Assistant Professor, EEE, UAP

6.8.26

Experiment No: 01

Experiment Name: Design and verification of RTL Logic Gate

Objective:

The objective of the experiment is

- To build and study logic gates: AND, OR, NOT, NAND or NOR from bipolar transistors using the RTL Logic family.
- To verify the truth table of constructed gates

Apparatus:

(1) Trainer Board ..... 01 Unit
(2) Digital Multi-meter..... 01 Unit
(3) Transistors ..... 03 Units
(4) Resistors ..... 05 Units
(5) Connecting wires ..... As needed

Circuit Diagram:

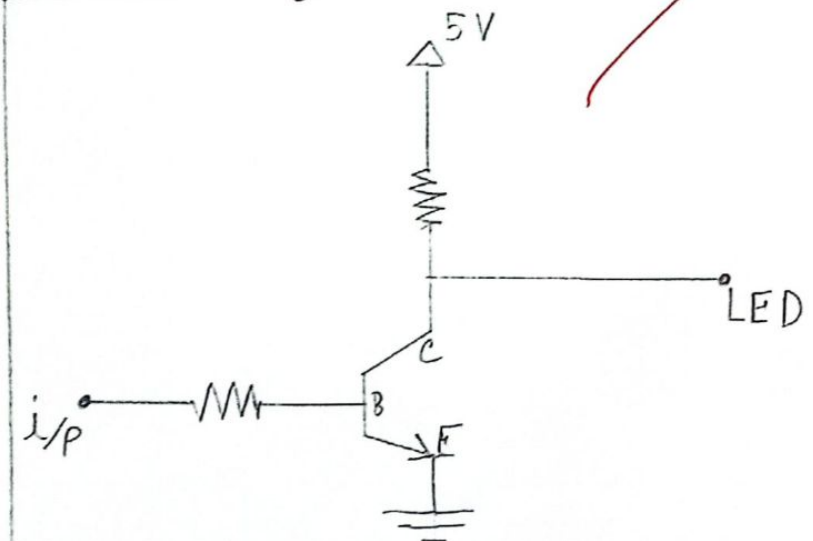


Figure 01: NOT Gate

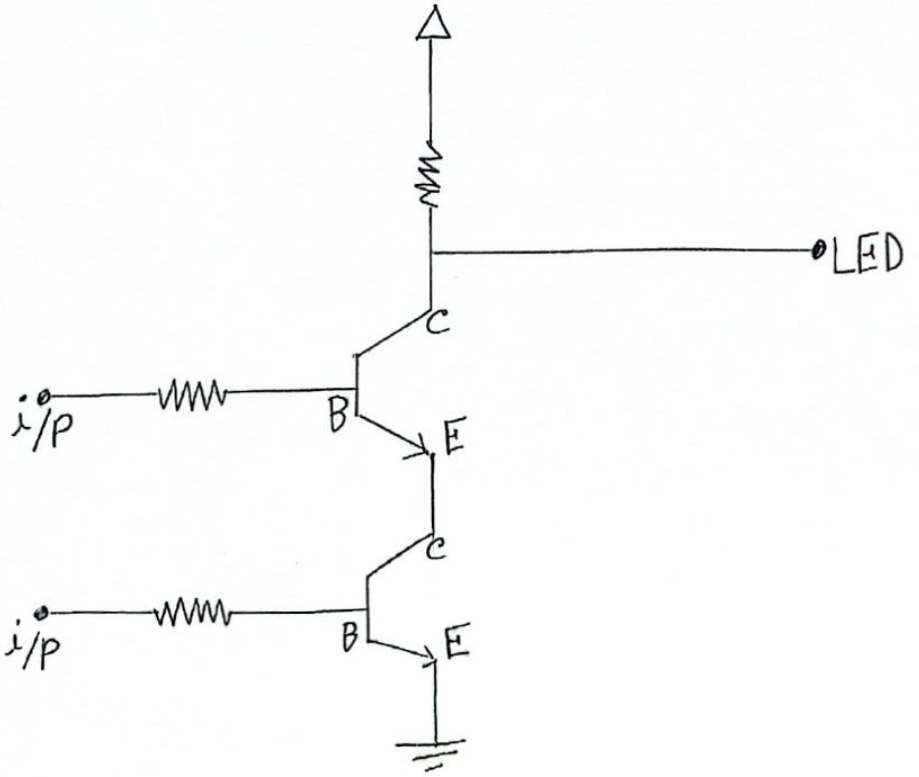


Figure 02: NAND Gate

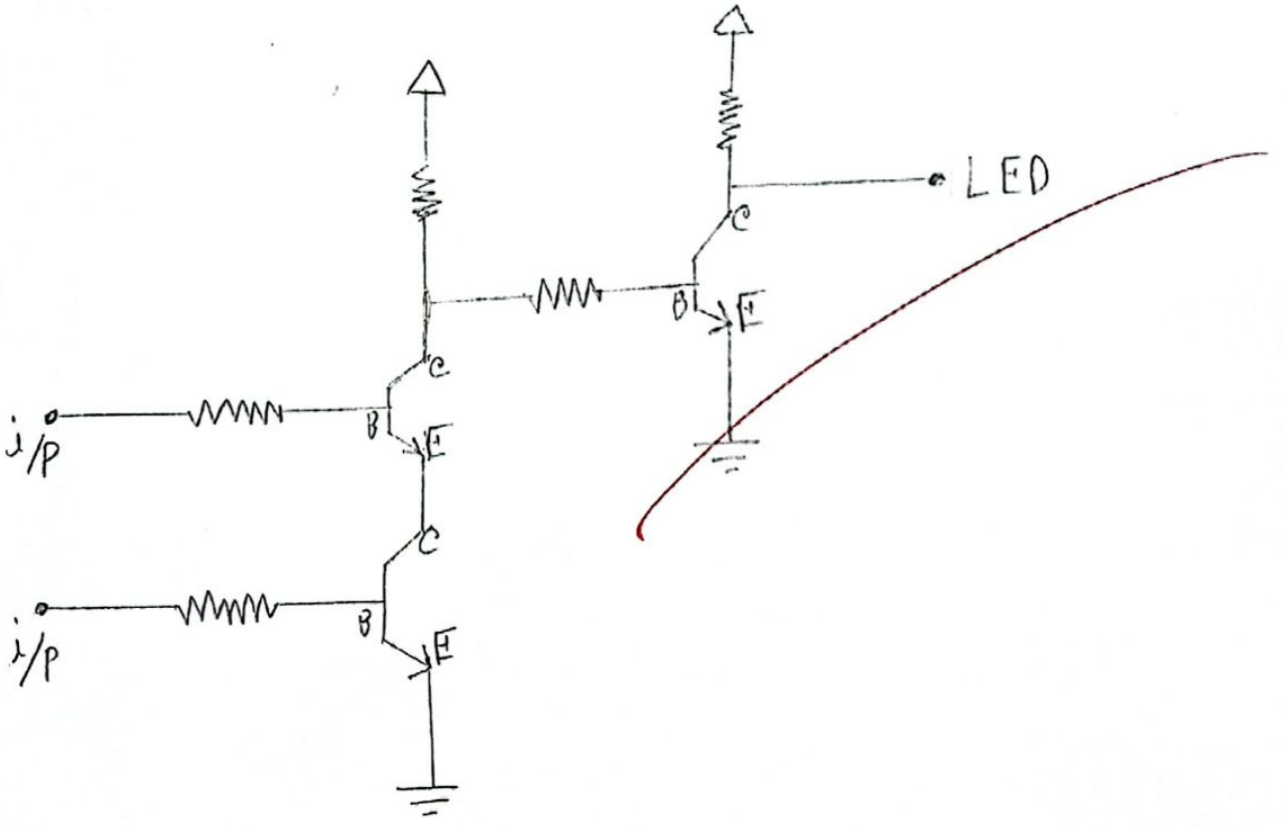


Figure 03: AND Gate

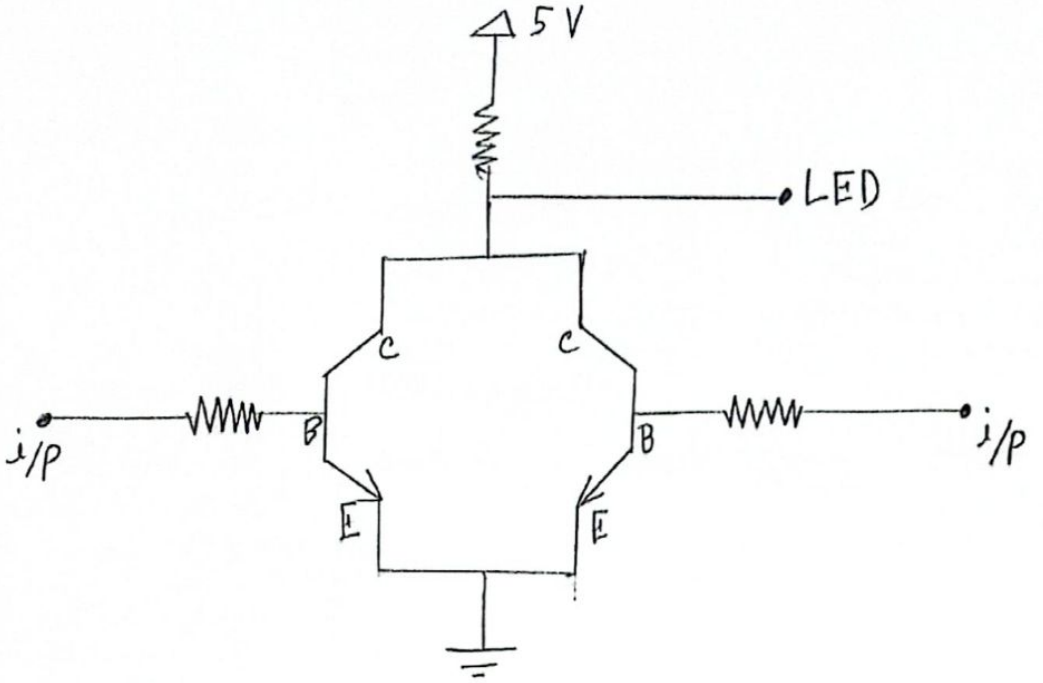


Figure 04: NOR Gate

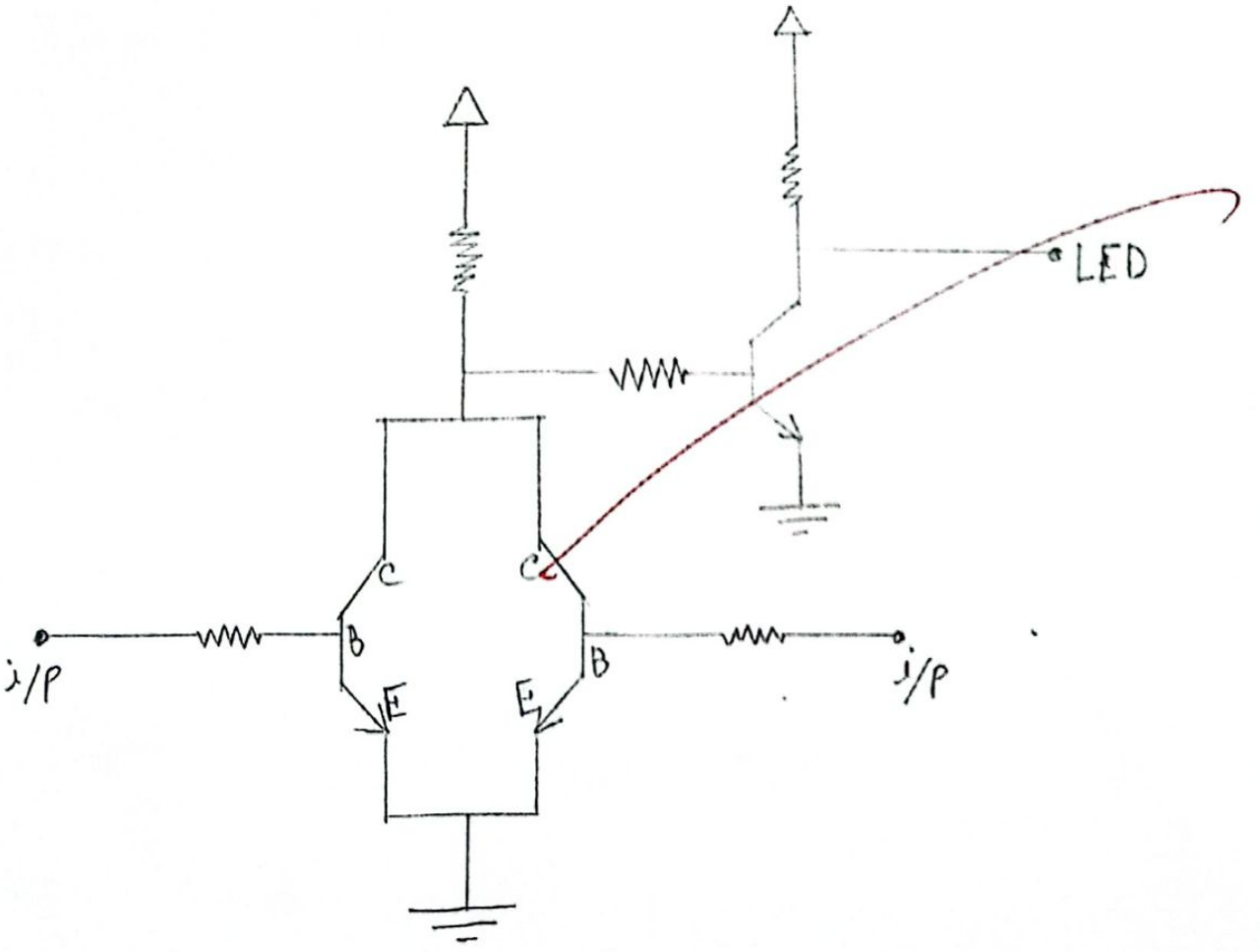


Figure 05: OR Gate

Data Table

a) Data for NOT Gate:

Input	Output Logical	Output Voltage
0	1	~~3.8~~ 3.70
1	0	0.08

b) Data for NAND Gate:

Input A	Input B	Output Logical	Output Voltage
0	0	1	3.67
0	1	1	3.82
1	0	1	3.66
1	1	0	0.44

c) Data for AND Gate:

Input A	Input B	Output Logical	Output Voltage
0	0	0	0.02
0	1	0	0.01
1	0	0	0.01
1	1	1	3.68

d) Data for NOR Gate:

Input A	Input B	Output Logical	Output Voltage
0	0	1	3.67 ~~0.05~~
0	1	0	0.01 ~~3.65~~
1	0	0	0.04 ~~3.67~~
1	1	0	0.03

e) Data for OR Gate:

Input A	Input B	Output Logical	Output Voltage
0	0	0	0.05
0	1	1	3.65
1	0	1	3.67
1	1	1	3.67

Discussion

In this experiment, we designed and verified basic RTL (Resistor-Transistor) Logic gates. We used transistors (BJT), resistors and measured output to verify the truth tables of logic gates (NOT, NAND, AND, NOR, OR). The logic operations were verified by observing the output voltages corresponding to different input combinations. The experiment successfully confirmed the theoretical behavior of these gates.

However, during the execution of the experiment, one of the transistors were burned out due to error in circuit connection. This incident highlighted the importance of doble-checking circuit connections before powering the setup.

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1: NOT Gate using RTL logic :-

Input	Output	V
0	1	3.70
1	0	0.08

2. NAND Gate using RTL logic :-

Input A	B	Output	V
0	0	1	3.67
0	1	1	3.82
1	0	1	3.66
1	1	0	0.04

3. AND Gate using RTL logic :-

A	B	Output	V
0	0	0	0.02
0	1	0	0.01
1	0	0	0.01
1	1	1	3.68

4. NOR Gate using RTL logic :

A	B	output	V
0	0	1	3.67
0	1	0	0.04
1	0	0	0.05
1	1	0	0.03

5. OR

A	B	output	V
0	0	0	0.05
0	1	1	3.65
1	0	1	3.67
1	1	1	3.67

University Of Asia Pacific

Department of Computer Science and Engineering (CSE)

Course Title: Electronic Devices and Circuits Lab

Course Code: EEE 202

No. of Experiment : 02

Name of Experiment : Design and Verification of DTL Logic Gates

Date of Experiment : 04/08/2026

Date of Submission : 11/08/2026

Submitted By

Name : Utsob Ghsoh

Registration No : 25101003

Section : A1

Submitted To

Abdullah Al Mahmud

Assistant Professor, EEE, UAP

11-8-26

Experiment No: 02

Experiment Name: Design and verification of DTL Logic gates

Objective:

The objective of this experiment is:

- To build and study logic gates: NOR, NOT, and OR, NAND and AND from bipolar transistors using the DTL logic family.
- To verify the truth table of constructed gates

Apparatus:

1. Trainer Board
2. Digital Multi-meter
3. Transistor (c-828)
4. Diode (1N4007)
5. Resistor (1K)
6. Connecting Wires

Circuit Diagram:

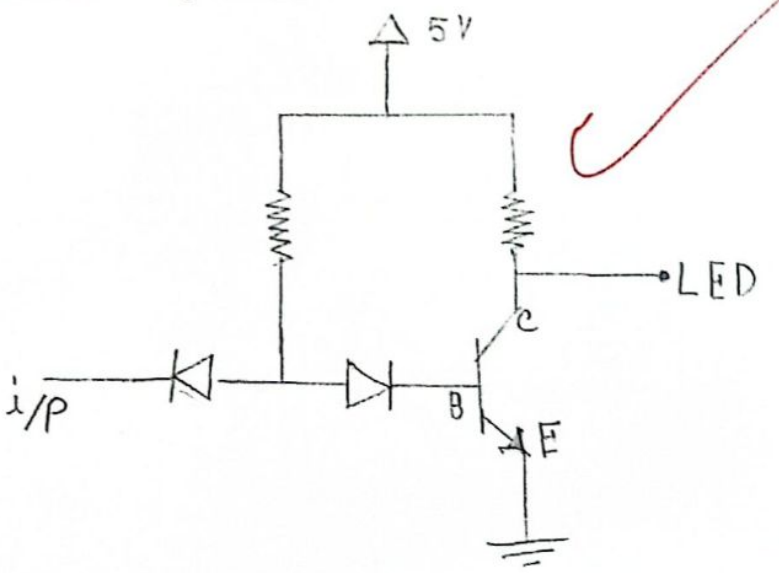


Figure 01:- NOT Gate using DTL Logic circuit

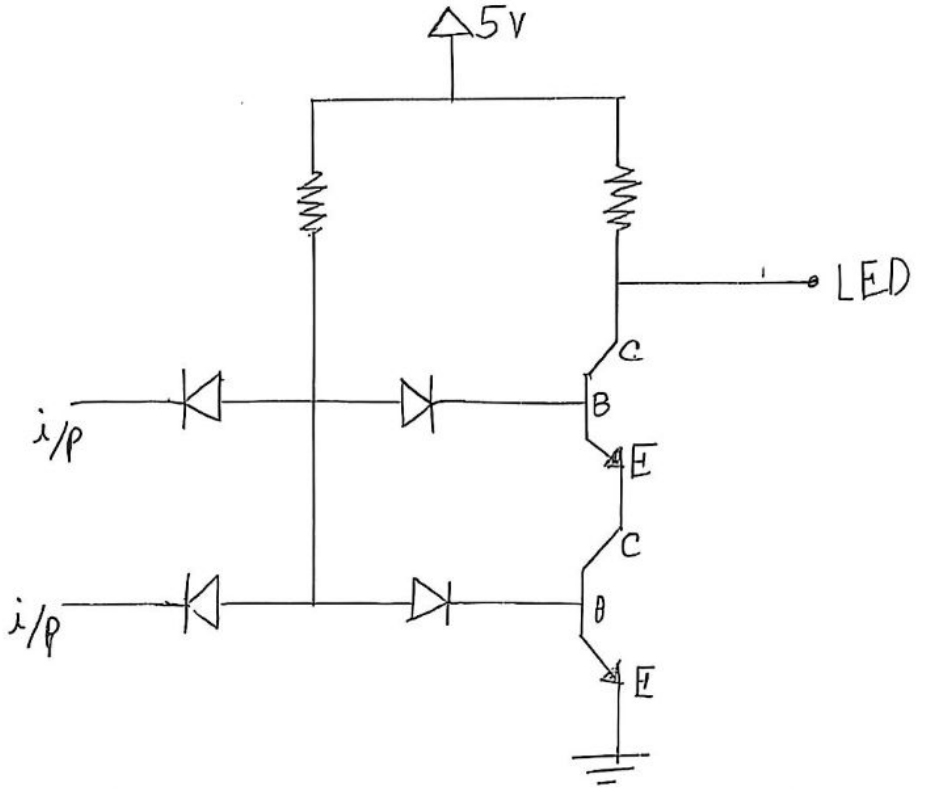


Figure 02:- NAND Gate using DTL Logic Circuit

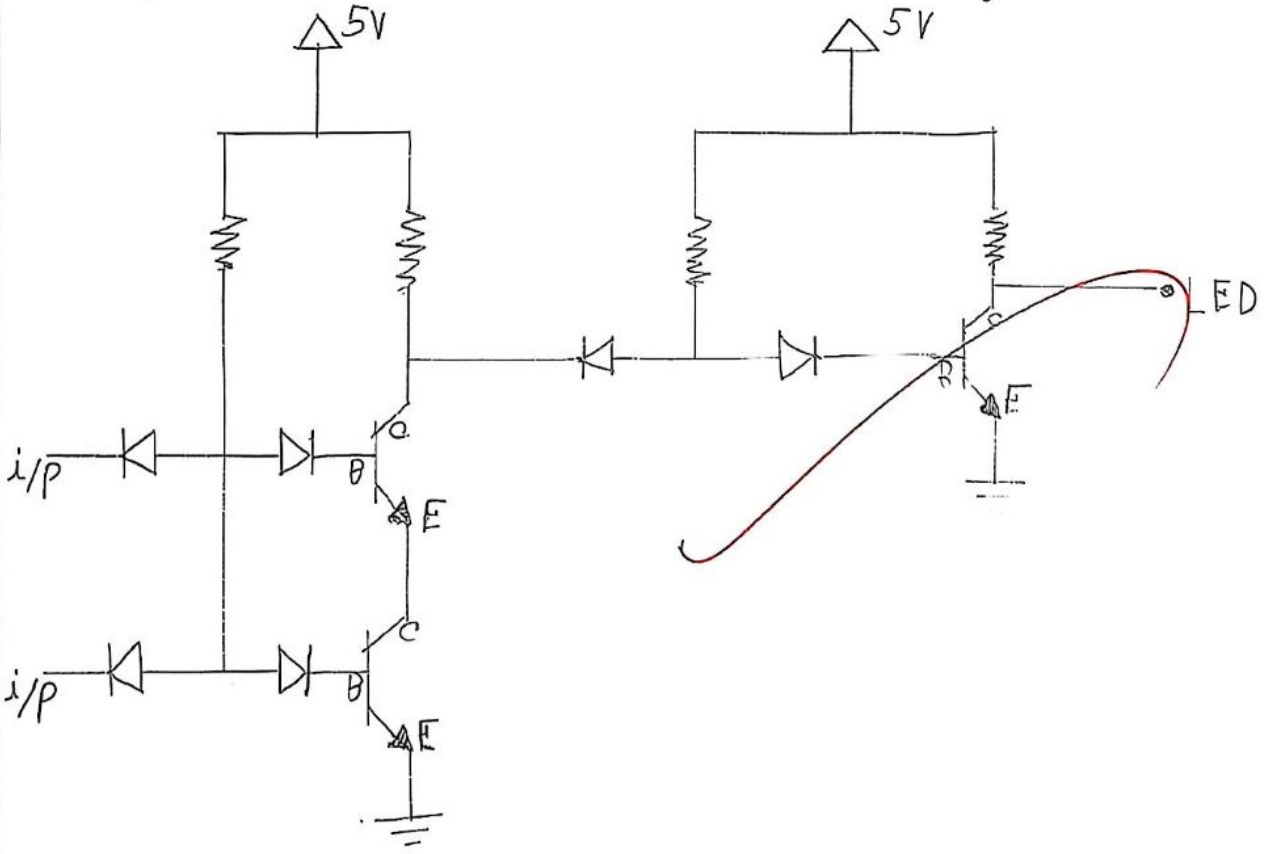


Figure 03:- AND Gate using DTL Logic Circuit

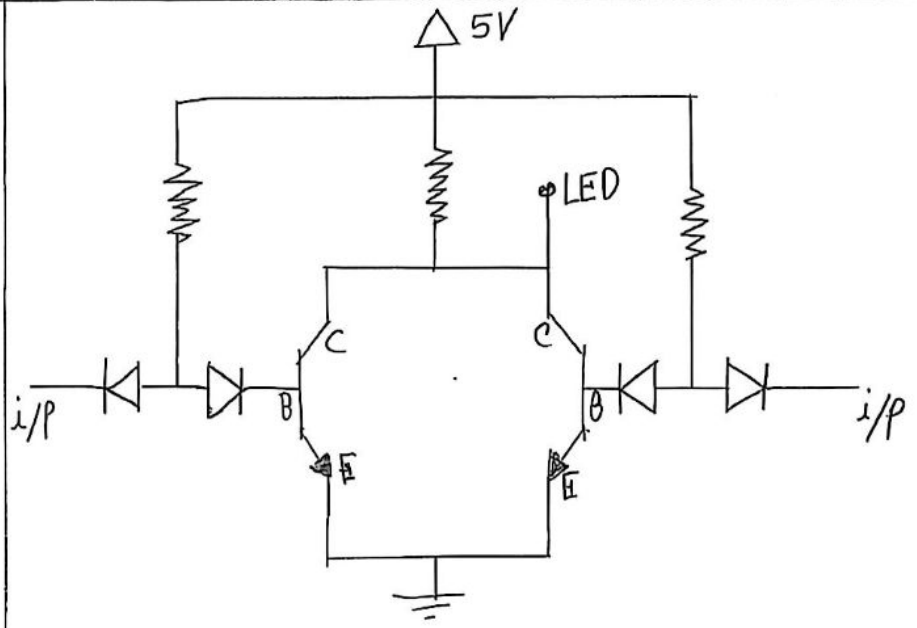


Figure 04:- NOR Gate using DTL Logic Circuit

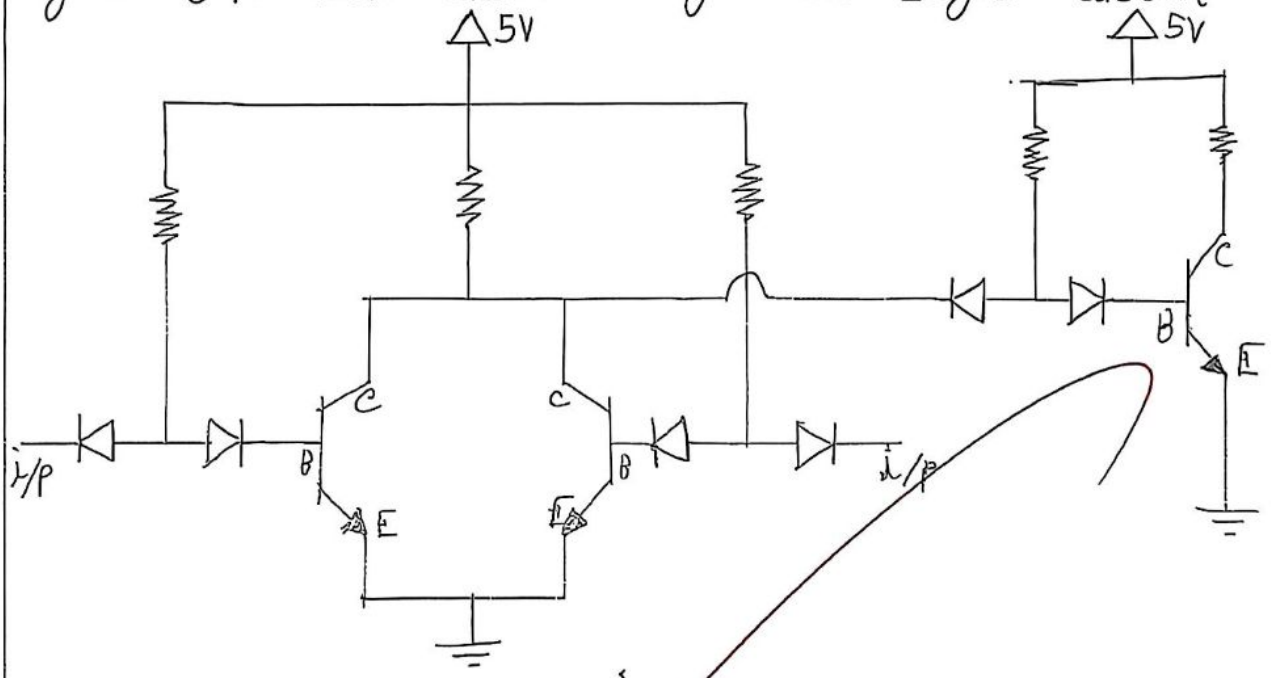


Figure 05:- OR Gate using DTL Logic Circuit

Data table:

a. Data for NOT Gate:

Input	Output Logical	Output Voltage
1	0	0.12
0	1	3.70

b. Data for NAND Gate:

Input A	Input B	Output Logical	Output Voltage
0	0	1	3.68
0	1	1	3.68
1	0	1	3.68
1	1	0	0.03

c. Data for NOR Gate:

Input A	Input B	Output Logical	Output Voltage
0	0	1	3.62
0	1	0	0.02
1	0	0	0.01
1	1	0	0.01

d. Data for AND Gate:

Input A	Input B	Output Logical	Output Voltage
0	0	0	0.12
0	1	0	0.10
1	0	0	0.10
1	1	1	3.61

e. Data for OR Gate:

Input A	Input B	Output Logical	Output Voltage
0	0	0	0.33
0	1	1	3.58
1	0	1	3.60
1	1	1	3.56

Discussion:

In this experiment, we successfully desined and verified the functionality of a specific DTL Logic gate. The design process involved using diodes for Logical AND operations and transistors for signal amplification. We applied different input combination using logic switches to observe and confirm the gates Logical behavior.

This experiment provided a better knowledge in understanding the operations of DTL gates, their advantages and limitations.

Data Table:

a. Data for NOT gate:

Input	output Logical	output voltage
1	0	0.12
0	1	3.70

b. Data for NAND gate:

Input A	Input B	Output Logical	Output voltage
0	0	1	3.868
0	1	1	3.68
1	0	1	3.68
1	1	0	0.03

c. Data for NOR gate:

Input A	Input B	Output Logical	Output voltage
0	0	1	3.62
0	1	0	0.02
1	0	0	0.01
1	1	0	0.01

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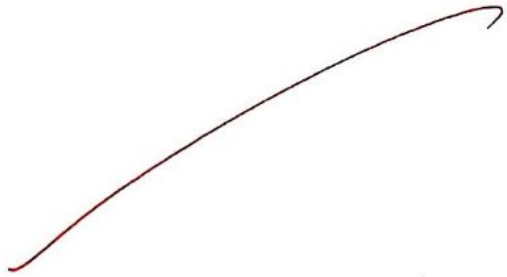
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d. Data for ~~AND~~ AND gate:

Input A	Input B	output Logical	output voltage
0	0	0	0.12
0	1	0	~~3.56~~ 0.1
1	0	0	0.10
1	1	1	3.61

e. Data for OR gate:

Input A	Input B	Output Logical	output voltage
0	0	0	0.33
0	1	1	3.58
1	0	1	3.60
1	1	1	3.56



University Of Asia Pacific

Department of Computer Science and Engineering (CSE)

Course Title: Electronic Devices and Circuits Lab

Course Code: EEE 202

No. of Experiment : 03

Name of Experiment : Analysis of Logic Gates using nMOS

Date of Experiment : 11/08/2026

Date of Submission : 18/08/2026

Submitted By

Name : Utsob Ghsoh

Registration No : 25101003

Section : A1

Submitted To

Abdullah Al Mahmud

Assistant Professor, EEE, UAP

18-8-26

Experiment No: 03

Experiment Name: Analysis of Logic Gates using NMOS

Objective:

The objective of the experiment is to analyze the MOSFET gate circuits and to understand the internal operation of the circuit. The truth table of these gates have to be verified by giving different logic level at the input

Apparatus:

1. nMos (IRF540/IRF740)
2. Resistors (1 kΩ)
3. Trainer Board
4. Multi-meter
5. Connecting Wires

Circuit Diagram:

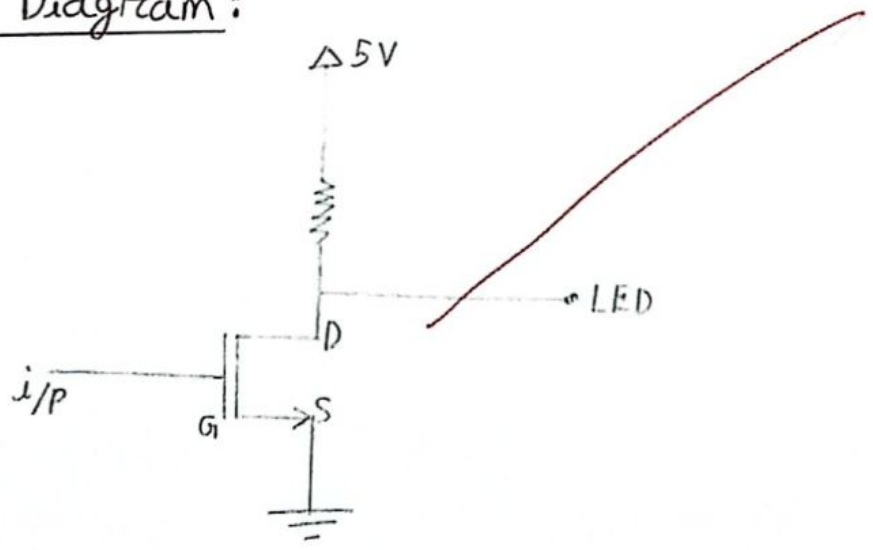


Figure 01: NMOS NOT Gate

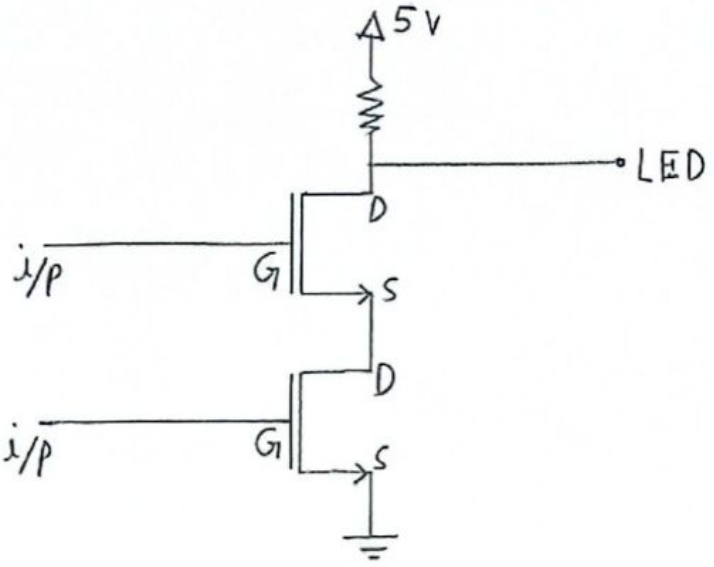


Figure 02 - NMOS NAND Gate

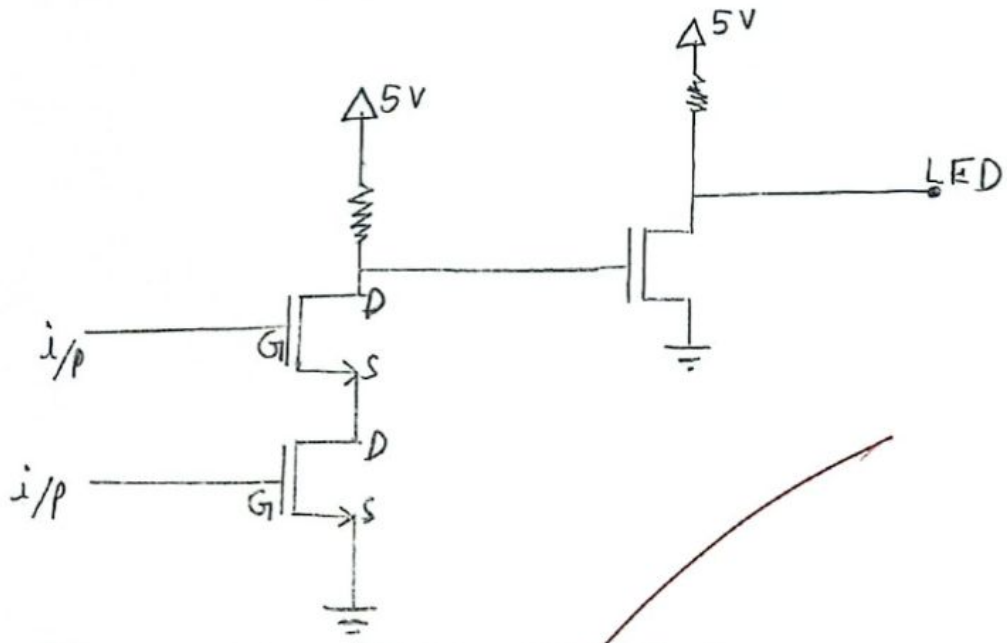


Figure 03 - NMOS AND Gate

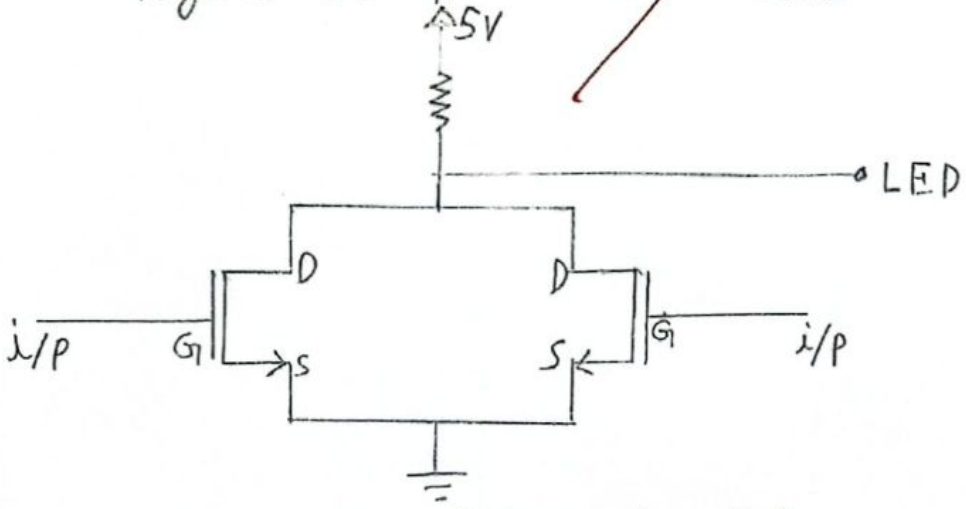


Figure 04 - NMOS NOR Gate

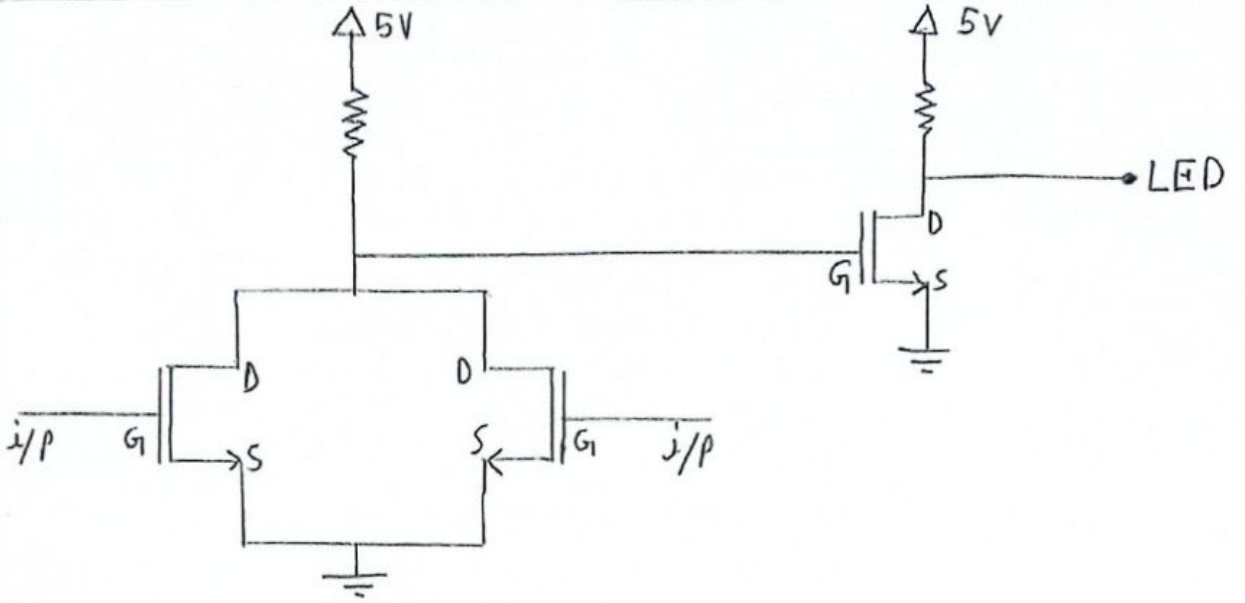


Figure 05 - NMOS OR Gate

Data Table:

a) Data for NOT gate:

Input	Output Logical	Output Voltage
0	1	3.68
1	0	0.01

b) Data for NAND Gate:

Input A	Input B	Output Logical	Output Voltage
0	0	1	3.66
0	1	1	3.66
1	0	1	3.66
1	1	0	0.08

c) Data for NOR gate:

Input A	Input B	Output Logical	Output Voltage
0	0	1	3.07
0	1	0	0.16
1	0	0	0.05
1	1	0	0.04

d) Data for AND gate:

Input A	Input B	Output Logical	Output Voltage
0	0	0	0.03
0	1	0	0.03
1	0	0	0.03
1	1	1	3.67

e) Data for OR gate:

Input A	Input B	Output Logical	Output Voltage
0	0	0	0.01
0	1	1	3.67
1	0	1	3.68
1	1	1	3.67

Discussion:

NMOS logic gates are constructed using n-type MOSFETs arranged in a pull-down network. The output is pulled high using either a resistor or a pull-up transistor. The design implements logic functions such as AND, OR, and NOT by controlling how the MOSFET's conduct based on input voltages.

To verify the gate, input values are applied, and the output is measured using a multimeter. The results are then compared with the expected truth table. Proper connection and power supply must be checked to ensure correct operation.

Data Table:

a. Data for NOT gate:

Input	output Logical	Output voltage
0	1	3.68
1	0	0.01

b. Data for NAND gate:

Input A	Input B	output Logical	Output voltage
0	0	1	3.66
0	1	1	3.66
1	0	1	3.66
1	1	0	0.08

c. Data for NOR gate:

Input A	Input B	output Logical	Output voltage
0	0	1	3.07
0	1	0	0.16
1	0	0	0.05
1	1	0	0.04

d. Data for AND gate:

Input A	Input B	output Logical	Output voltage
0	0	0	0.03
0	1	0	0.03
1	0	0	0.03
1	1	1	3.67

e. Data for OR gate:

Input A	Input B	output Logical	output voltage
0	0	0	0.01
0	1	1	3.67
1	0	1	3.68
1	1	1	3.67

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University Of Asia Pacific

Department of Computer Science and Engineering (CSE)

Course Title: Electronic Devices and Circuits Lab

Course Code: EEE 202

No. of Experiment : 04

Name of Experiment : Design and Verification of TTL Logic Gates

Date of Experiment : 18/08/2026

Date of Submission : 25/08/2026

Submitted By

Name : Utsob Ghsoh

Registration No : 25101003

Section : A1

Submitted To

Abdullah Al Mahmud

Assistant Professor, EEE, UAP

Experiment NO: 04

Experiment Name: Design and Varification of TTL logic Gates

Objective:

The objective of this experiment is:

- To study NOT, NAND, AND, NOR and OR gates using the TTL logic circuit.
- To build NOT, NAND, AND, NOR and OR using bipolar transistors.
- To verify the truth table of constructed gates.

Diagram:

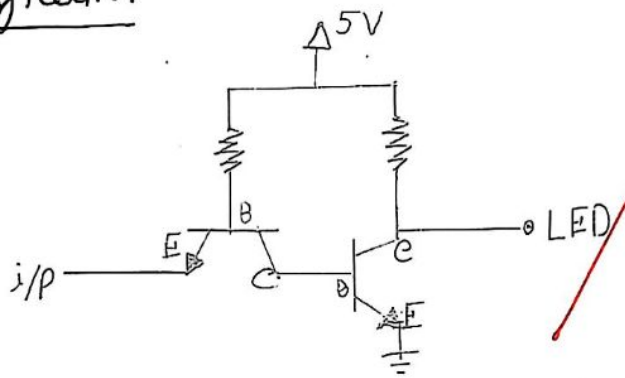


Figure 1: NOT Gate using TTL Logic Circuit

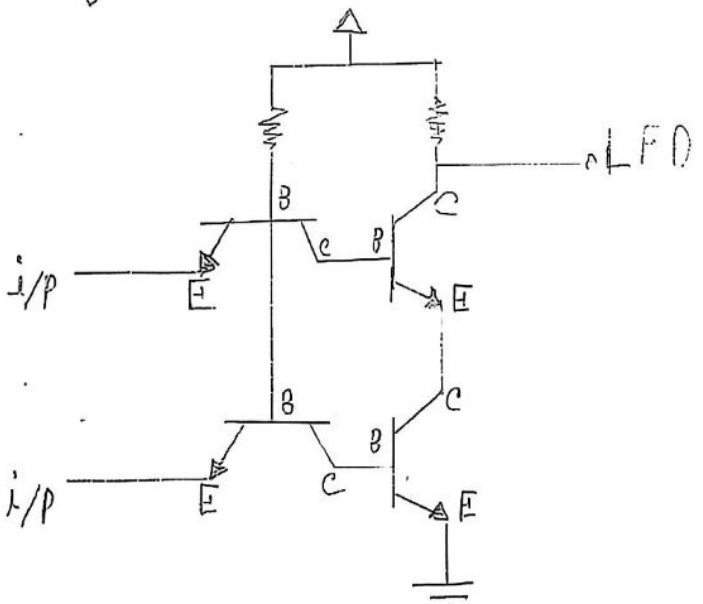


Figure 2: NAND Gate using TTL Logic Circuit

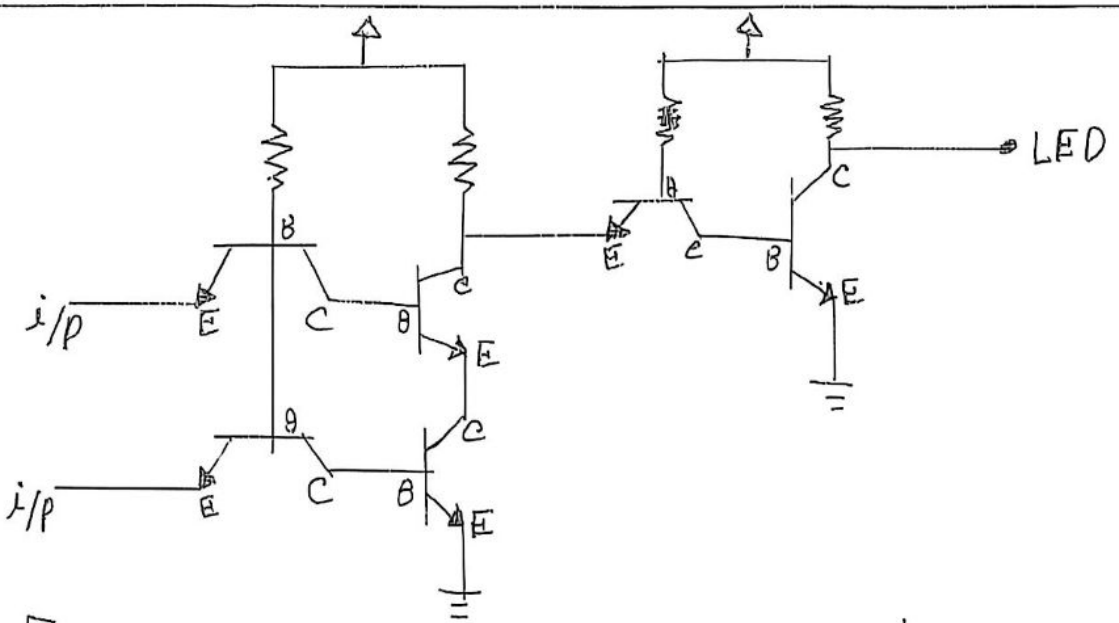


Figure 3: AND Gate using TTL Logic Circuit

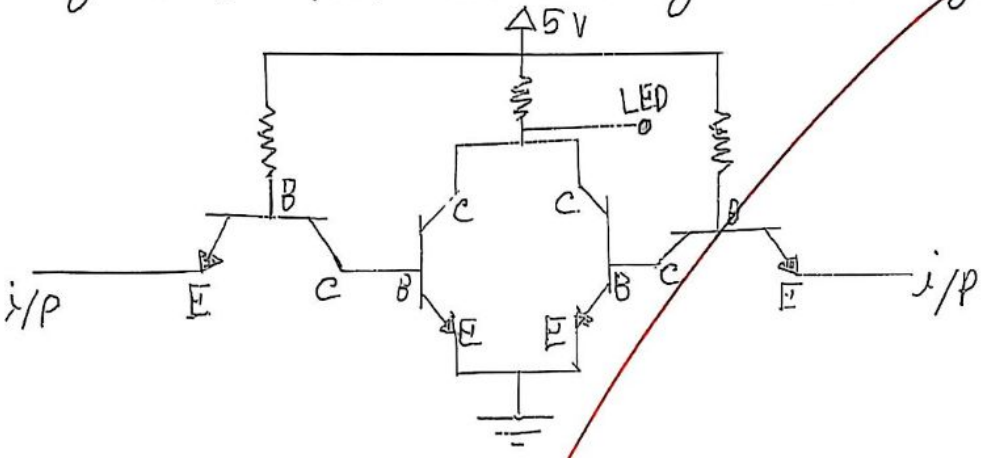


Figure 4: NOR Gate using TTL Logic Circuit

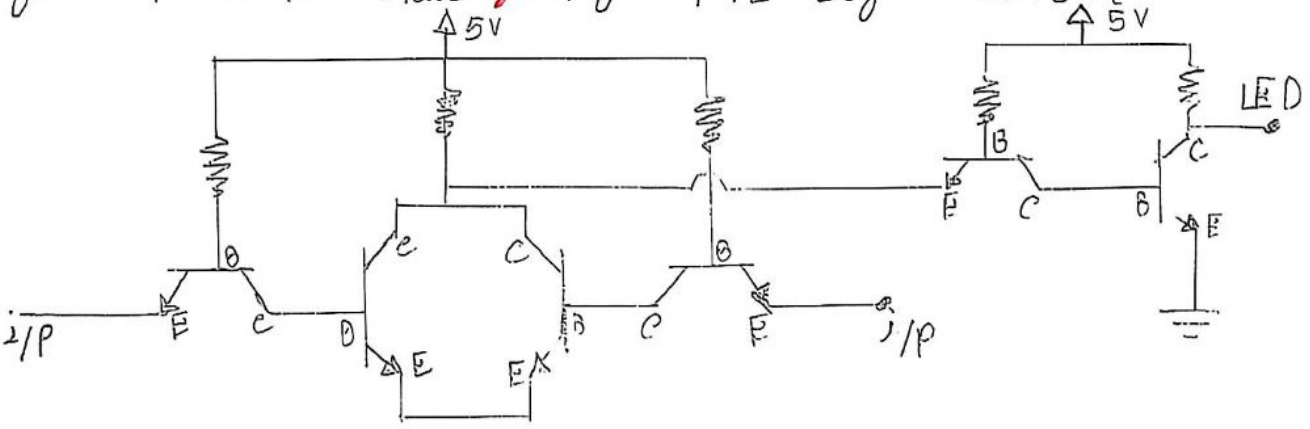


Figure 5: OR Gate using TTL Logic circuit

Data Table:

Data for NOT gate:

Input	Output Logical	Output Voltage
0	1	3.66
1	0	0.07

Data for NAND gate:

Input A	Input B	Output Logical	Output Voltage
0	0	1	3.64
0	1	1	3.64
1	0	1	3.63
1	1	0	0.09

Data for NOR gate:

Input A	Input B	Output Logical	Output Voltage
0	0	1	3.64
0	1	0	0.09
1	0	0	0.04
1	1	0	0.06

Data for AND gate:

Input A	Input B	Output Logical	Output Voltage
0	0	0	0.29
0	1	0	0.24
1	0	0	0.19
1	1	1	3.65

Data for OR gate:

Input A	Input B	Output Logical	Output Voltage
0	0	0	0.03
0	1	1	3.67
1	0	1	3.67
1	1	1	3.68

Discussion:

The objective of this experiment was to design and verify the operation of basic TTL Logic gates (AND, OR, NOT, NAND, NOR) and to confirm their truth tables through practical observation.

During the experiment, the circuits were constructed on a breadboard and all possible input combinations were applied to each gate. The output was monitored using LED's to represent logical on(1) and off(0).

This confirmed that the desired result was achieved as the practical outputs successfully verified the expected logical behaviour of the TTL gates.

Overall, the experiment provided a clear understanding of how TTL Logic gates operate and demonstrated the reliability of theoretical logic principles in practical operations.

Data Table

Data for NOT gate

Input	output Logical	output voltage
0	1	3.66
1	0	0.07

Data for NAND gate

Input A	Input B	output Logical	output voltage
0	0	1	3.64
0	1	1	3.64
1	0	1	3.63
1	1	0	0.09

Data for NOR gate

Input A	Input B	Output Logical	output voltage
0	0	1	3.64
0	1	0	0.09
1	0	0	0.04
1	1	0	0.06

Data for AND gate

Input A	Input B	output Logical	output voltage
0	0	0	0.29
0	1	0	0.24
1	0	0	0.19
1	1	1	3.65

Data for OR gate

Input A	Input B	Output Logical	output voltage
0	0	0	0.03
0	1	1	3.67
1	0	1	3.67
1	1	1	3.68

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University Of Asia Pacific

Department of Computer Science and Engineering (CSE)

Course Title: Electronic Devices and Circuits Lab

Course Code: EEE 202

No. of Experiment : 05

Name of Experiment : Analysis of Logic Gates using CMOS

Date of Experiment : 25/08/2026

Date of Submission : 01/09/2026

Submitted By

Name : Utsob Ghsoh

Registration No : 25101003

Section : A1

1.9.26

Submitted To

Abdullah Al Mahmud

Assistant Professor, EEE, UAP

Experiment No : 05

Experiment Name: Analysis of logic Gates using CMOS

Objective:

The objective of the experiment is to analyze the MOSFET gate circuits and to understand the internal operation of the circuit. The truth table of these gates have to be verified by giving different logic level at the input.

Apparatus:

1. nMOS
2. pMOS
3. Trainer Board
4. Multi_Meter
5. Connecting Wires

Circuit Diagram:

NOT gate:

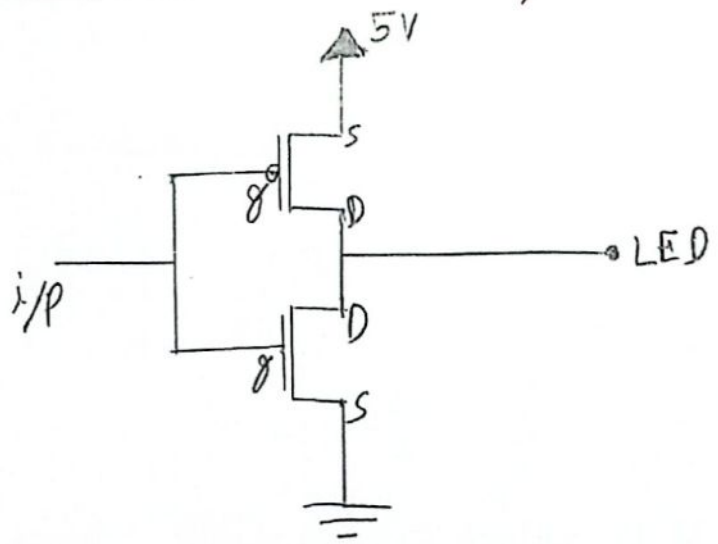


Figure: NOT gate using CMOS

NAND gate:

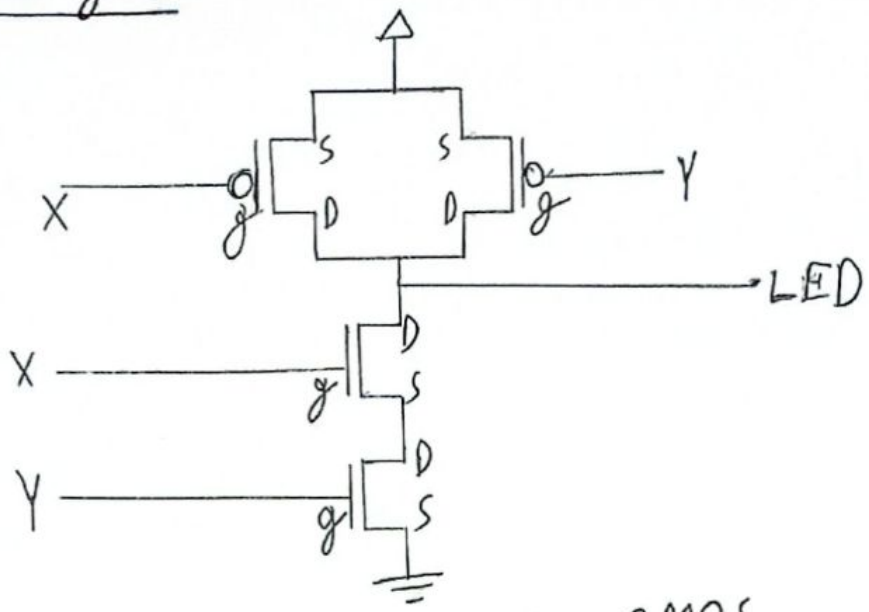


Figure: NAND gate using CMOS

AND gate:

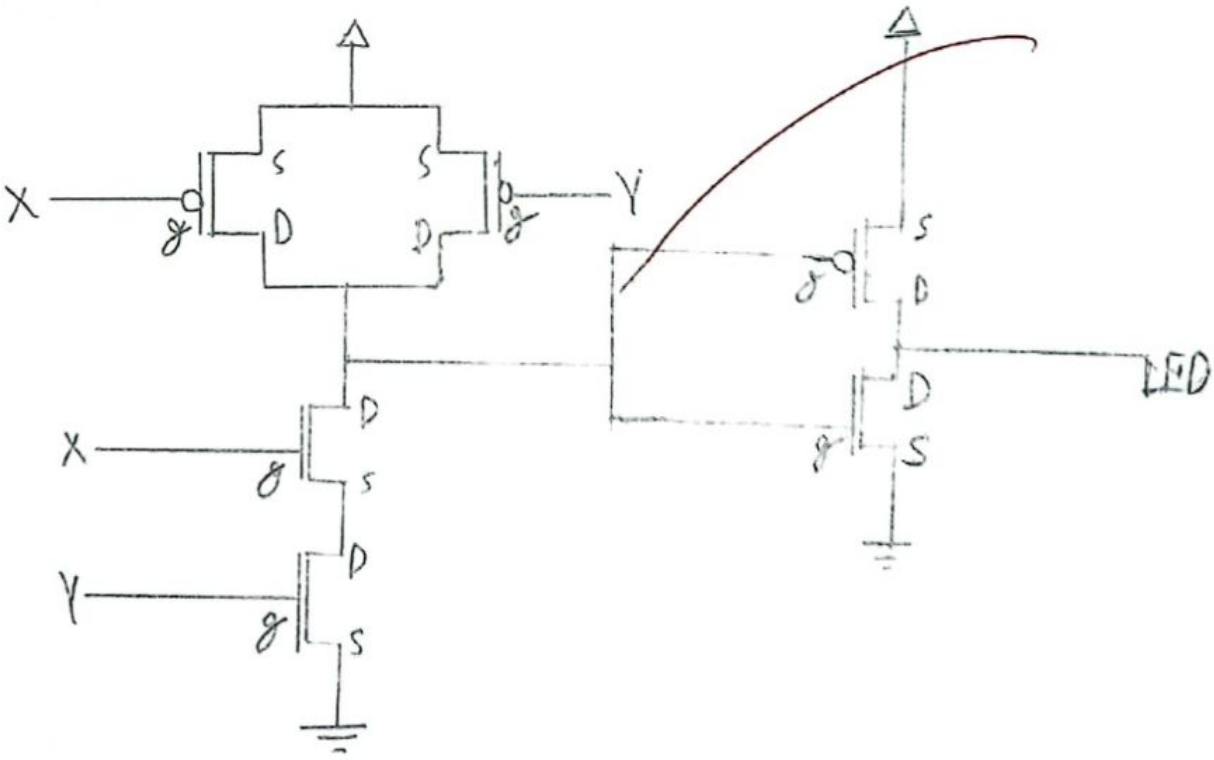


Figure: AND gate using CMOS

NOR gate:

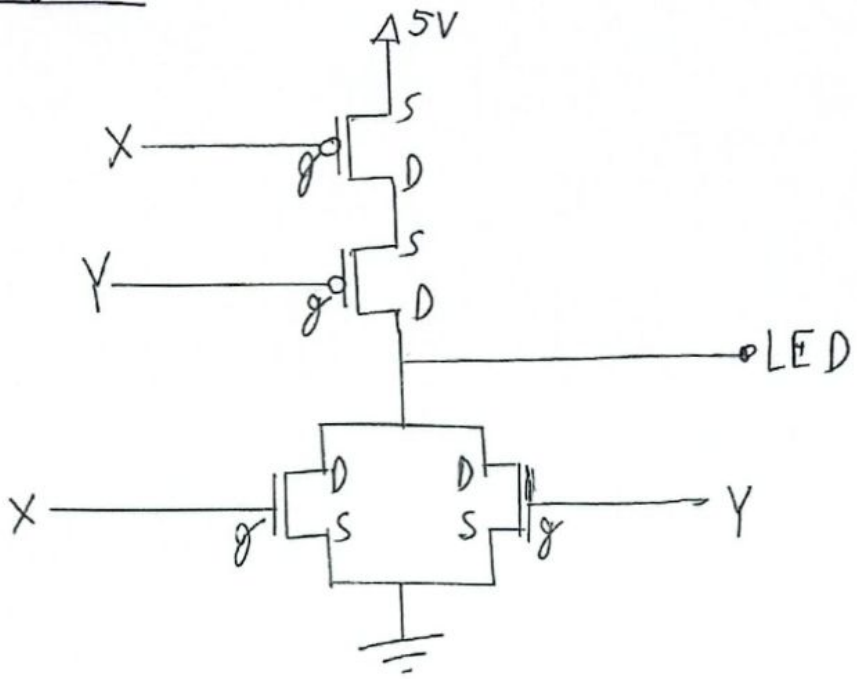
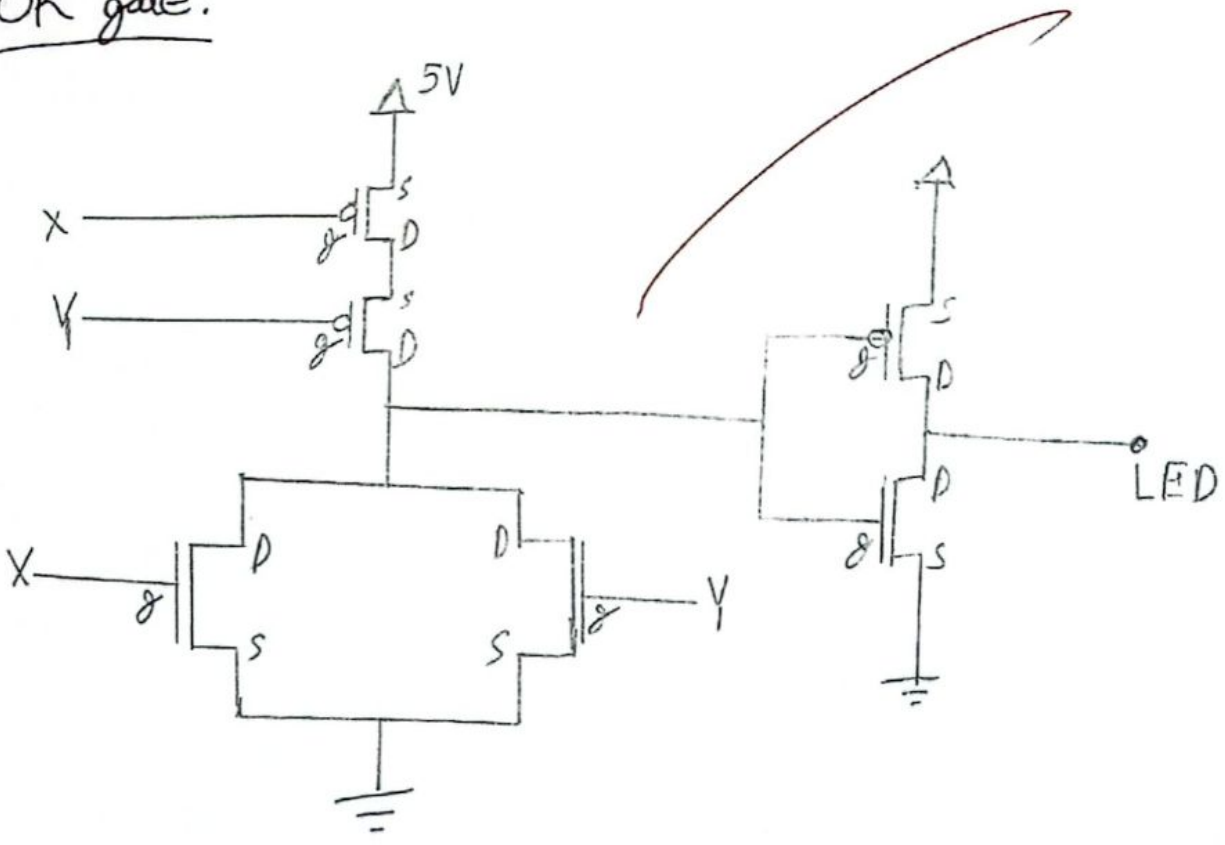


Figure: NOR gate using CMOS

OR gate:



Data Table:

a. NOT Gate:

Input	Output Logical	Output Voltage
0	1	5.04
1	0	~~0.06~~ 0.01

b. NAND Gate:

Input A	Input B	Output Logical	Output Voltage
0	0	1	5.04
0	1	1	5.18
1	0	1	5.20
1	1	0	0.01

c. AND Gate;

Input A	Input B	Output Logical	Output Voltage
0	0	0	0.00
0	1	0	0.00
1	0	0	0.00
1	1	1	5.05

d. NOR Gate:

Input A	Input B	Output Logical	Output Voltage
0	0	1	5.04
0	1	0	0.00
1	0	0	0.00
1	1	0	0.00

e. OR Gate:

Input A	Input B	Output Logical	Output Voltage
0	0	0	0.00
0	1	1	5.05
1	0	1	5.05
1	1	1	5.05

Discussion:

In this experiment, CMOS logic gates such as NOT, NAND, AND, NOR, OR were designed and tested. The observed outputs matched the expected truth tables, which verified the current operation of the CMOS gates.

The experiment also highlighted the key advantages of CMOS technology. The gates consumed very little static power, ~~po produced~~ produced sharp transitions in the output and provided full logic voltage levels with high noise immunity.

Data Table

Data for NOT gate

Input	Output Logical	Output Voltage
0	1	5.04
1	0	0.01

Data for NAND gate

Input A	Input B	Output Logical	Output Voltage
0	0	1	5.04
0	1	1	5.18
1	0	1	5.20
1	1	0	0.01

Data for NOR gate

Input A	Input B	Output Logical	Output Voltage
0	0	1	5.04
0	1	0	0.00
1	0	0	0.00
1	1	0	0.00

Data for AND gate

Input A	Input B	Output Logical	Output Voltage
0	0	0	0.00
0	1	0	0.00
1	0	0	0.00
1	1	1	5.05

Data for OR gate

Input A	Input B	Output Logical	Output Voltage
0	0	0	0.00
0	1	1	5.05
1	0	1	5.05
1	1	1	5.05

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University Of Asia Pacific

Department of Computer Science and Engineering (CSE)

Course Title: Electronic Devices and Circuits Lab

Course Code: EEE 202

No. of Experiment : 06

Name of Experiment : Study of a Single Phase Transformer and Determination of Turn Ratio

Date of Experiment : 01/09/2026

Date of Submission : 08/09/2026

Submitted By

Name : Utsob Ghsoh

Registration No : 25101003

Section : A1

8-9-26

Submitted To

Abdullah Al Mahmud

Assistant Professor, EEE, UAP

Experiment No : 06

Experiment Name: study of single phase transformer and Determination of turn ratio

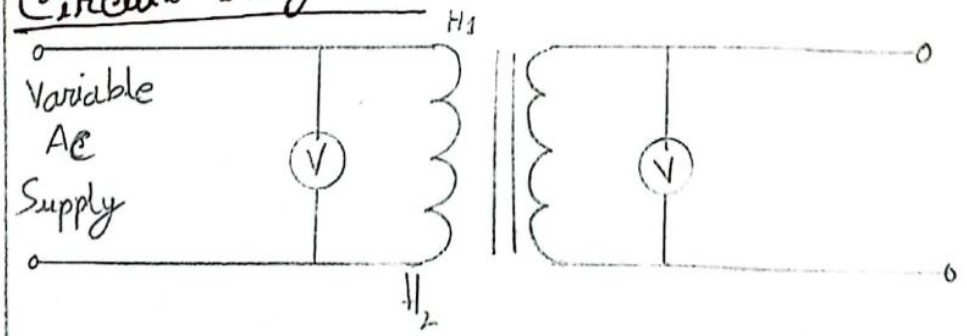
Objective:

To study the construction and working principle of a single-phase transformer and to determine its turn ratio by measuring the voltages on the primary and secondary windings.

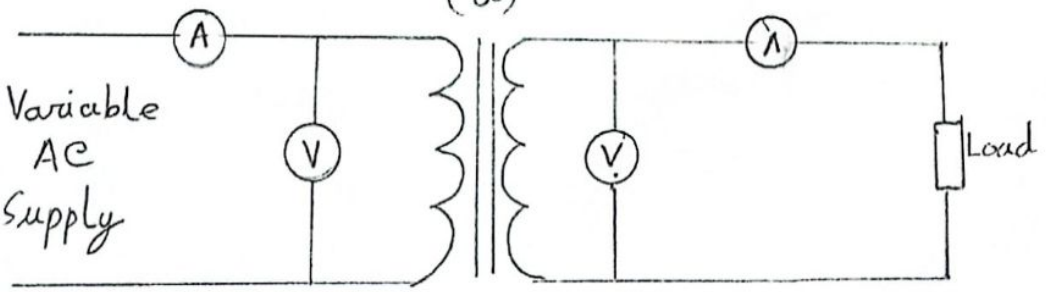
Apparatus:

1. Two AC voltmeters (0-300v)
2. Two AC ammeter (0-10A)
3. One single phase transformer
4. One rheostat (Current rating greater than or ~~equal~~ equal to 5A)
5. Auto transformer (Variac)
6. Wires and Cords.

Circuit Diagram:



(a)



(b)

Data Table:

A. Voltage Ratio;-

E_1 (V)	E_2 (V)	$a = \frac{E_1}{E_2}$
40.3 V	20.1	2.004
81.3 V	40.6	2.002
120.8 V	60.3	2.003
162.2 V	80.5	2.002

B. Current Ratio:

E_1 (V)	I_1 (A)	I_2 (A)	$a = \frac{I_2}{I_1}$
40.3 V	0.05	0.10	2.0
81.3 V	0.10	0.19	1.9
120.8 V	0.15	0.27	1.8
161.2 V	0.20	0.33	1.65

Discussion:

A single phase transformer works on electro-magnetic induction. The voltage ratio between the primary and secondary is proportional to the turns ratio. In this experiment, the measured voltage were used to calculate the turns ratio. The result was close to the theoretical value, with small differences caused by core losses, leakage flux and measurement errors.

Data table:

A. Voltage Ratio:

$E_1(V)$	$E_2(V)$	$a = \frac{E_1}{E_2}$
40.3V	20.1	2.004
81.3V	40.6	2.002
120.8V	60.3	2.003
161.2 V	80.5	2.002

B. Current Ratio:

$E_1(V)$	$I_1(A)$	$I_2(A)$	$a = \frac{I_2}{I_1}$
40.3V	0.05	0.10	2.0
81.3V	0.10	0.19	1.9
120.8V	0.15	0.27	1.8
161.2V	0.20	0.33	1.65

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University Of Asia Pacific

Department of Computer Science and Engineering (CSE)

Course Title: Electronic Devices and Circuits Lab

Course Code: EEE 202

No. of Experiment : 07

Name of Experiment : Study of Open Circuit Test and Short Circuit Test of a Single Phase Transformer

Date of Experiment : 08/09/2026

Date of Submission : 29/09/2026

Submitted By

Name : Utsob Ghsoh

Registration No : 25101003

Section : A1

29-9-26

Submitted To

Abdullah Al Mahmud

Assistant Professor, EEE, UAP

Experiment No: 07

Experiment Name: Study of open circuit Test and short circuit Test of a single phase Transformer

objective:

The objective of this experiment is to determine the efficiency, voltage regulation and the equivalent circuit parameters of a single phase transformer by performing:

1. Open circuit Test
2. Short circuit Test

Apparatus:

1. Single Phase Transformer
2. Voltmeter (AC)
3. Ammeter (AC)
4. Watt meter
5. Connecting wires
6. Auto transformer

Circuit Diagram:

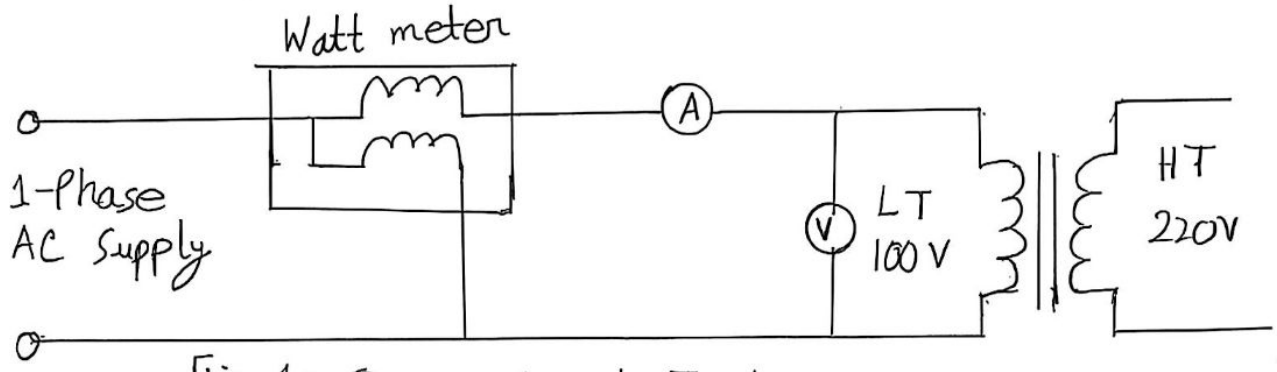


Fig 1: Open circuit Test circuit

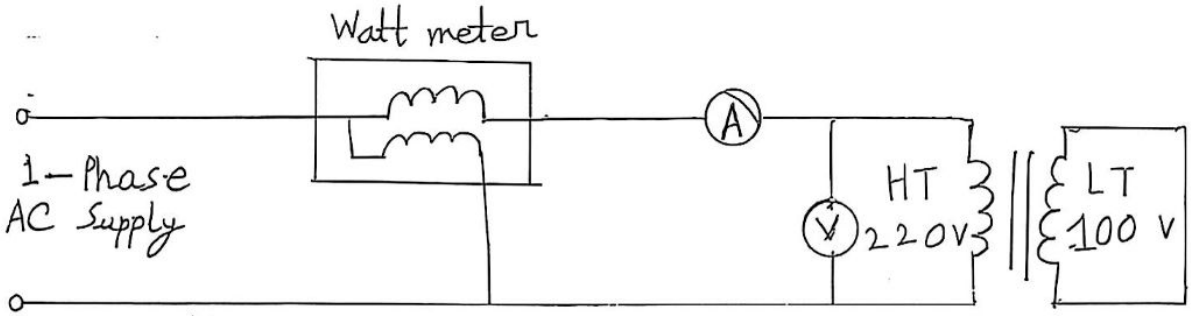


Fig 2: Short circuit Test circuit

Data Table:

Open circuit Test:

V_o(V)	I_o (mA)	P_o (W)	$\cos\theta_o = \frac{P_o}{V_o I_o}$	$I_w = I_o \cos\theta$	$I_u = I_o \sin\theta$
110.2	80.5 mA	3	0.338	27.2 mA	75.76 mA

Short circuit Test:

V(V)	I(A)	P(W)	$R_{01} = \frac{P_{sc}}{I^2_{sc}}$	$X_{01} = \sqrt{Z^2 - R_{01}^2}$
14.6	1.05	15.5	14.06	4.47i

Calculation:

$$\cos\theta = \frac{P_o}{V_o I_o} = \frac{3}{110.2 \times 0.0805} = 0.338$$

$$\sin\theta = \sqrt{1 - \cos^2\theta} = \sqrt{1 - (0.338)^2} = 0.9411$$

$$I_w = I_o \cos\theta = 80.5 \times 0.338 = 27.2 \text{ mA}$$

$$I_u = I_o \sin\theta = 80.5 \times 0.9411 = 75.76 \text{ mA}$$

$$R_{01} = \frac{P_{sc}}{I^2_{sc}} = \frac{15.5}{(1.05)^2} = 14.06$$

$$Z_{01} = \frac{V_{sc}}{I_{sc}} = 13.9$$

$$\therefore X_{01} = \sqrt{Z_{01}^2 - R_{01}^2} = \sqrt{13.9^2 - 14.06^2} = \sqrt{-4.47} = 4.47j$$

Discussion:

This experiment examined open-circuit and short circuit tests of a single phase transformer. The open circuit test measured core losses, while the short-circuit test determined copper losses and impedence due to high current. The short circuit test was brief to avoid overheating. These tests help evaluate effeciency and voltage regulation without load.

Open Circuit Test:
Data Sheet:

V_o (V)	I_o (mA)	P_o (W)	$\cos\theta = \frac{P}{V_o I_o}$	$I_w = I_o \cos\theta$	$I_\mu = I_o \sin\theta$
110.2	80.5 mA	3	0.93	80.49	0.474

Short Circuit Test
Data Sheet:

V(V)	I(mA)	P(W)	$R_o = \frac{P_{sc}}{I_{sc}}$	$X_{o1} = \sqrt{Z^2 - R_{o1}^2}$
14.6	1.05	15.5	14.76	$\sqrt{4.47}$

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University Of Asia Pacific

Department of Computer Science and Engineering (CSE)

Course Title: Electronic Devices and Circuits Lab

Course Code: EEE 202

No. of Experiment : 08

Name of Experiment : Study of Basic Configuration of Operational Amplifier

Date of Experiment : 29/09/2026

Date of Submission : 06/10/2026

Submitted By

Name : Utsob Ghsoh

Registration No : 25101003

Section : A1

Submitted To

Abdullah Al Mahmud

Assistant Professor, EEE, UAP

6.10.26

Exp No : 08

Experiment Name: Study of Basic Configuration of Operational Amplifier

Objective:

The objective of the experiment is to analyze the AC and DC performance of operational amplifier.

Apparatus:

(i) Op-amp IC 741 (2)
(ii) Resistor – 1 kΩ (2), 2 kΩ (2), 4 kΩ (2)
(iii) Single Generator
(iv) DC voltage source
(v) Trainer board
(vi) Osoilloscope
(vii) Multimeter

Circuit Diagram:

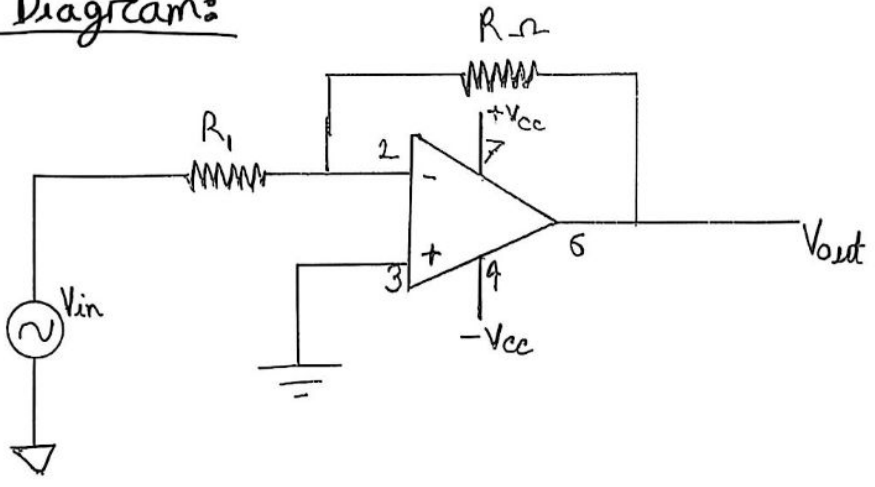


Fig 1: Inverting Amplifier

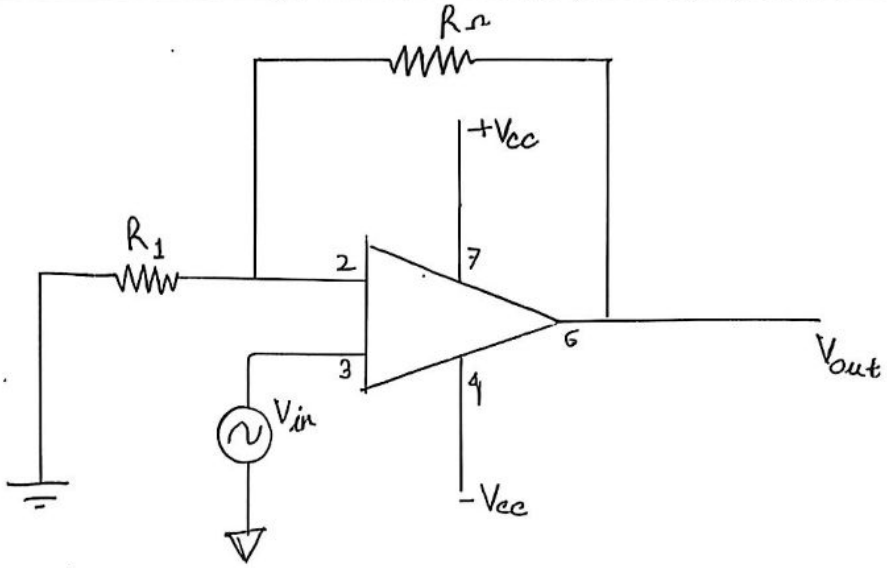


Fig 2 - Non-inverting Amplifier

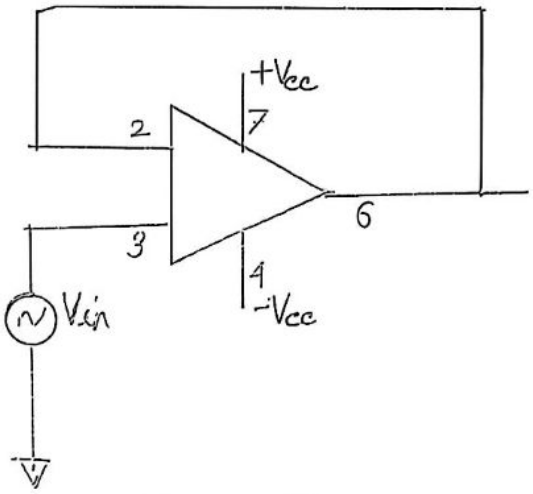


Figure 3 - Voltage Follower Circuit

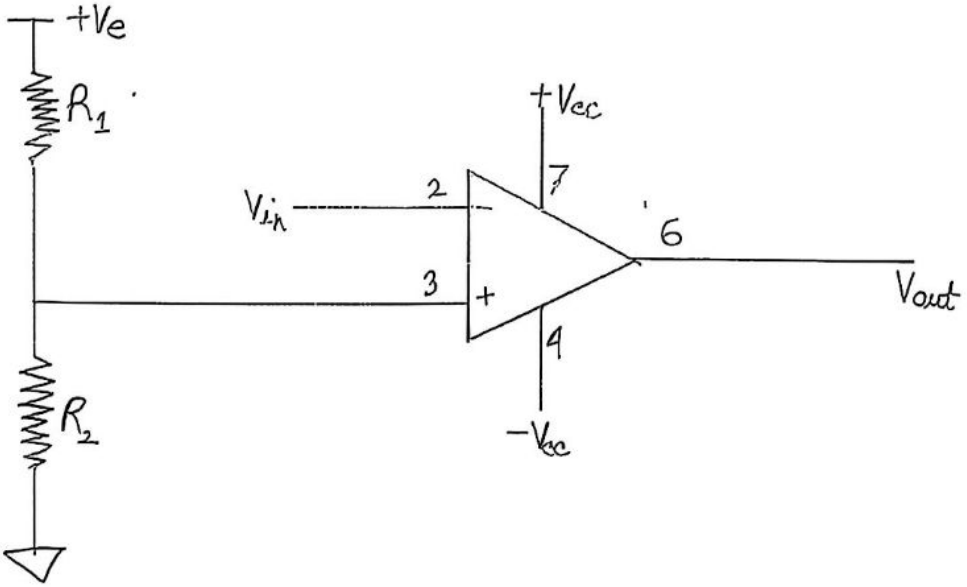
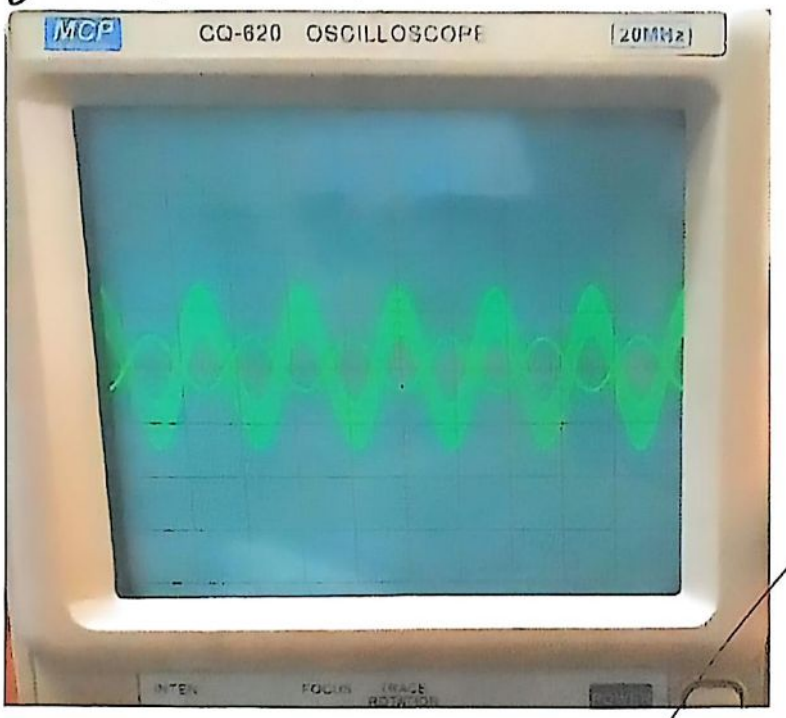


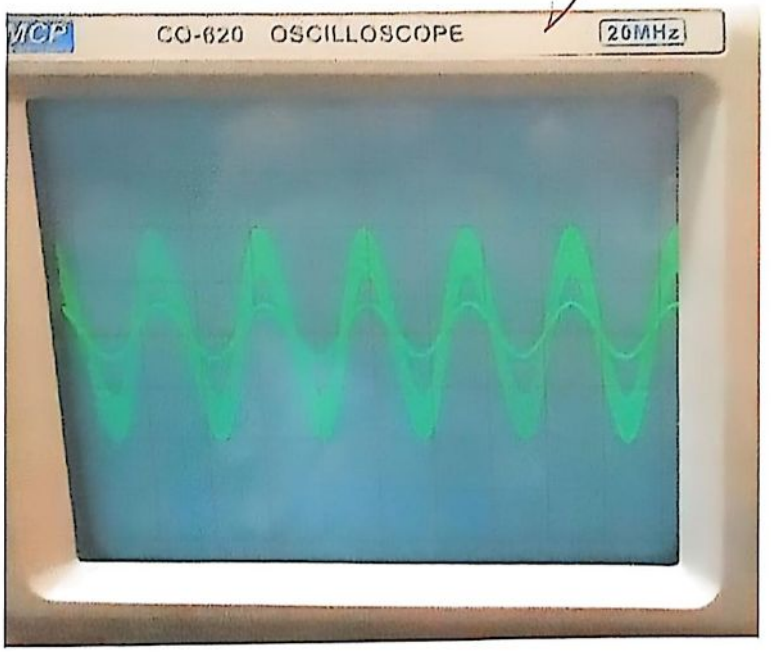
Figure 4 - Comparator circuit

Experimental Data:

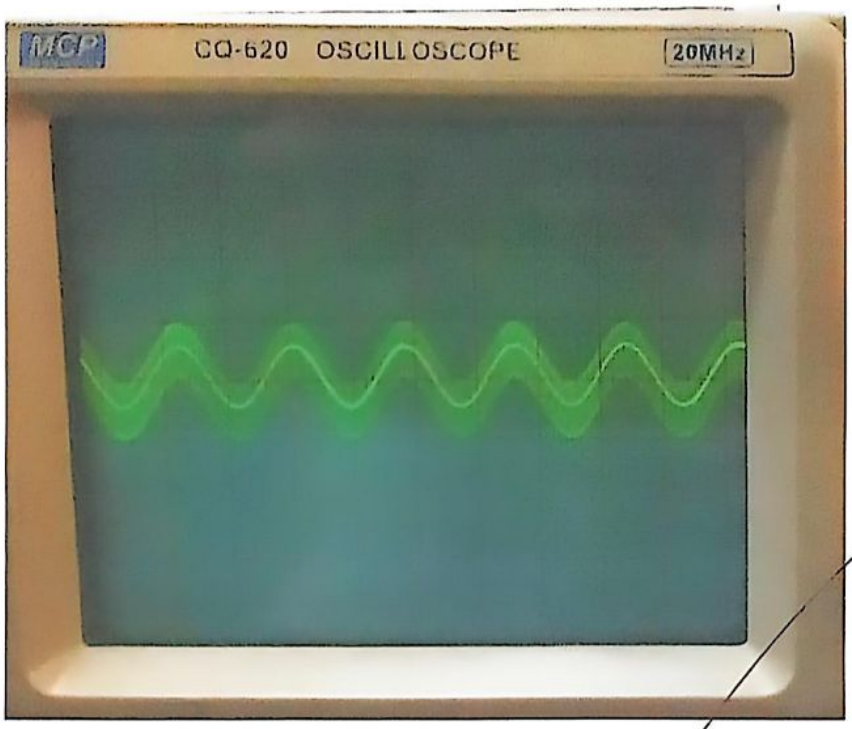
Inverting Amplifier input and output:



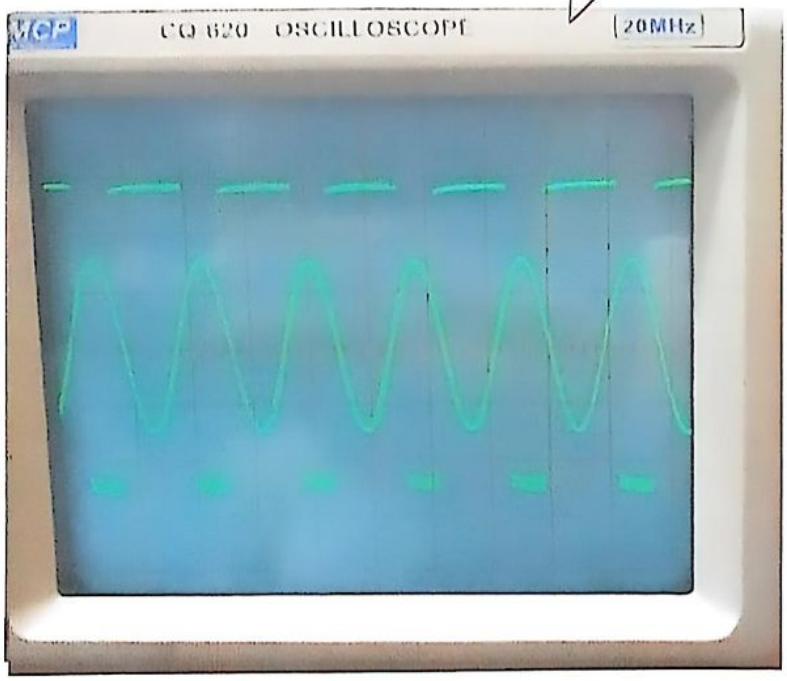
Non-inverting Amplifier input and output:



Voltage follower circuit input and output:



Comparator Circuit Input and Output:



Calculation:

Inverting Amplifier:

$$R_f = 2.16\ k\Omega$$

$$R_1 = 0.98\ k\Omega$$

$$V_o = -\left(\frac{R_f}{R_1}\right) * V_{in}$$

$$= -\left(\frac{216}{0.98}\right) * 1\ V_{p-p}$$

$$= -2.20\ V_{p-p}$$

Non-inverting Amplifier:

$$V_o = \left(1 + \frac{R_f}{R_1}\right) * V_{in}$$

$$= \left(1 + \frac{2.16}{0.98}\right) * 1\ V_{p-p}$$

$$= 3.20\ V_{p-p}$$

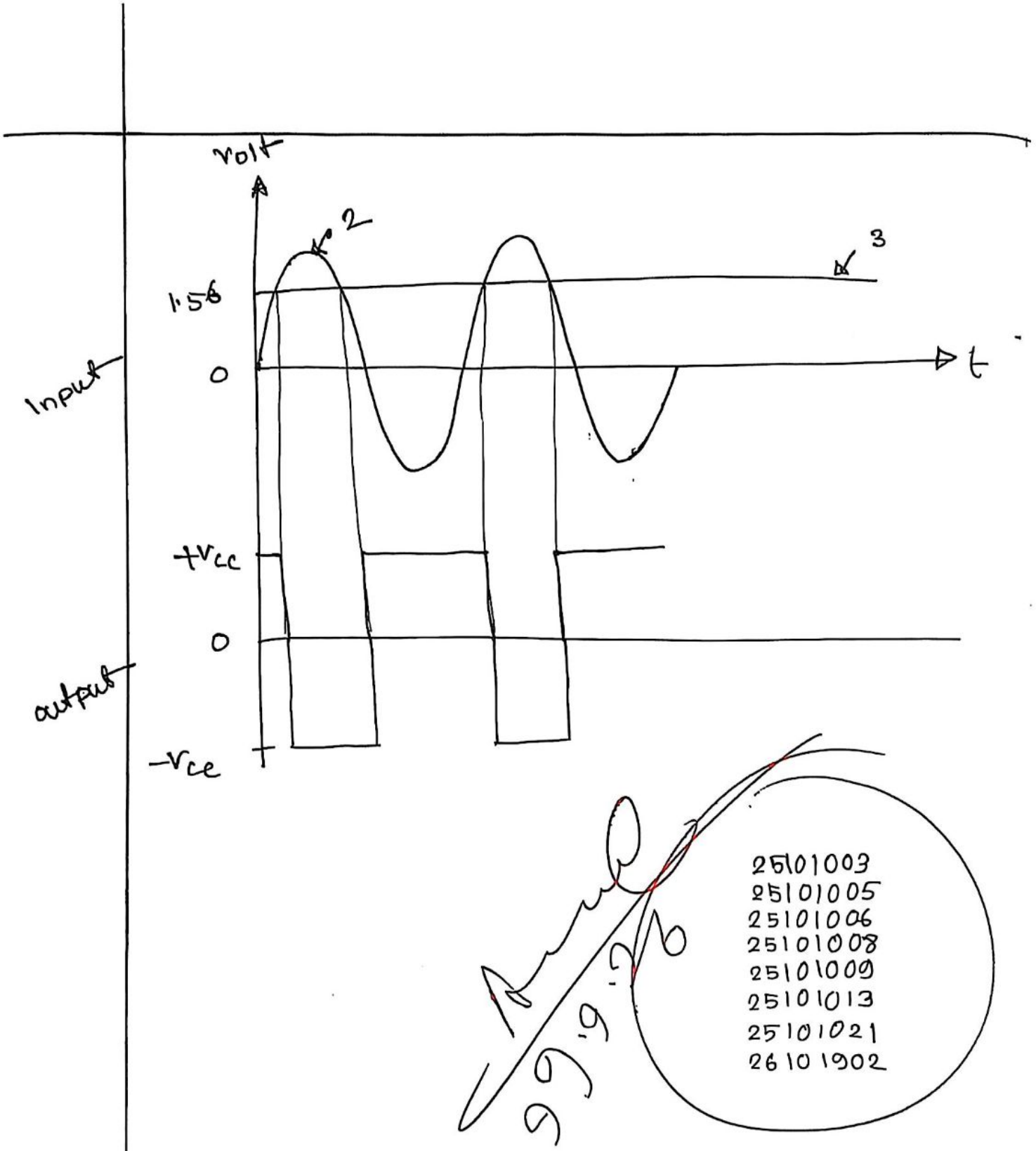
Discussion:

Through this experiment, we designed and analyzed different kinds of operational amplifiers like inverting amplifier, non-inverting amplifier, voltage follower circuit, comparator circuit. The amplifiers were built using op-amps and resistors, and the outputs of each circuit were using a multimeter and oscilloscope.

The results of the experimental data were found to be quite consistent with the theoretical values.

Apart from a few minor circuit issues, the experiment was set up properly after rechecking for any loose connections or misplacements.

In conclution, the experiment was sucessful and provided accurate results.


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2
3
1.58
0
t
Input
+Vcc
0
output
-Vce

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V = 15.03

V- = 15.00

Inverting Amp

$R_f = 2.16\ K\Omega$

$R_1 = 0.98\ K\Omega$

$$V_o = \left(\frac{R_f}{R_1}\right) * V_{in}$$

$$= -\left(\frac{2.16}{0.98}\right) * 1\ V_{p-p}$$

$$= -2.20\ V_{p-p} \quad (\text{theo})$$

$$V_o = -2.20\ V_{p-p} \quad (\text{Exp})$$

Non- inverting Amp.

$$V_o = \left(1 + \frac{R_f}{R_1}\right) * V_{in}$$

$$= \left(1 + \frac{2.16}{0.98}\right) * 1\ V_{p-p}$$

$$= 3.20\ V_{p-p} \quad (\text{theo.})$$

$$V_o = 3.2\ V_{p-p}. \quad (\text{Exp})$$

University Of Asia Pacific

Department of Computer Science and Engineering (CSE)

Course Title: Electronic Devices and Circuits Lab

Course Code: EEE 202

No. of Experiment : 09

Name of Experiment : Analysis of Binary Weighted D/A Converter

Date of Experiment : 06/10/2026

Date of Submission : 13/10/2026

Submitted By

Name : Utsob Ghsoh

Registration No : 25101003

Section : A1

Submitted To

Abdullah Al Mahmud

Assistant Professor, EEE, UAP

University Of Asia Pacific

Department of Computer Science and Engineering (CSE)

Course Title: Electronic Devices and Circuits Lab

Course Code: EEE 202

No. of Experiment : 10

Name of Experiment : Analysis of Astable Multivibrator using 555 timer

Date of Experiment : 06/10/2026

Date of Submission : 13/10/2026

Submitted By

Name : Utsob Ghsoh

Registration No : 25101003

Section : A1

Submitted To

Abdullah Al Mahmud

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